

Use Case: Suppose You have created local repository and you want to push that repository on github.

If you have created a repository on local system and you want to push that repository on github then follow the below steps.

- Step 1: Go to your workspace folder (Devops)
- Step 2: Create a new folder (GIT-PRACTICE) and open git bash inside it.
- Step 3: Initialize GIT_PRACTICE as an empty git repository by using "git init" command
- Step 3: Create 1 empty file say T1.txt
- Step 4: Add T1.txt into staging area
- Step 5: Add T2.txt into commit area.
- Step 6: try to push this repository on git hub by using "git push" command, but you will get an error

```
$ git push
fatal: No configured push destination.
Either specify the URL from the command-line or configure a remote repository using

    git remote add <name> <url>

and then push using the remote name

    git push <name>
```

Reason for this error is git unable to find the destination, how to resolve this error?

Solution: First create same repository on GitHub (GIT_PRACTICE).

We need to create a link between local repository and central repository.

To create a link between local repository and central repository use below command.

Syntax: `git remote add origin central_repository_url`

Here remote- manages remote on Local repository

add -To add new remote to your list of remote

origin-It is Remote name, we can use any name but origin is most commonly used.

This setting is required only one time

Now try push the local repository changes into github by using "git push" command

Still, we will get an error

```
$ git push
fatal: The current branch master has no upstream branch.
To push the current branch and set the remote as upstream, use

    git push --set-upstream origin master

To have this happen automatically for branches without a tracking upstream, see 'push.autoSetupRemote' in 'git help config'.
```

Reason for this error is -we are pushing local repository changes to central repository so we need to specify upstream branch. To set upstream branch use below command

Syntax:

```
git push -u origin master
```

Here -u is used for Tracking set up.

Tracking set up - It set up tracking relationship between your local master branch and central repository master branch.

Once you set up upstream branch then next time onward you can run git push and git pull command directly on that repository.

Note: If local repository root branch name and central repository root branch name is not same then we need to rename local repository root branch name.

To rename local repository branch name use below command

Syntax: `git branch -M master` (If Central repository root branch name is master)

`git branch -M main` (If Central repository root branch name is main)

i.e If we are pushing local repository changes to central then local repository branch and central repository branch name should be same, if not same then rename local repository branch same as to central repository name by using above command.